Radians

1

a 0.56 **b** 0.83 **c** 0.83 **d** 1.48

e -4.38 **f** -0.50 **g** 0.87 **h** -0.81

i -0.59

Exact values

2

a $h = 1$ unit

b **i** $\frac{\sqrt{3}}{2}$ **ii** $\frac{1}{2}$ **iii** $\sqrt{3}$ **iv** $\frac{1}{2}$

3

a $h = \sqrt{2}$ units

b **i** $\frac{\sqrt{2}}{2}$ **ii** $\frac{\sqrt{2}}{2}$ **iii** 1

4

a 1 **b** 0 **c** 0 **d** 1

e 0

5

a $\frac{\sqrt{3}+1}{2}$ **b** $\frac{\sqrt{2}}{4}$ **c** 1 **d** $\frac{\sqrt{6}}{4}+1$

e 0 **f** 2

Exact values from reference angles

6 $\alpha = \frac{\pi}{11}$

7 $\alpha = \frac{\pi}{7}$

8 $\alpha = \frac{\pi}{5}$

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9

a $\frac{1}{2}$ b -1 c $-\frac{1}{2}$ d $-\frac{\sqrt{3}}{2}$

e $-\frac{\sqrt{3}}{2}$ f $\frac{1}{2}$ g 1 h 0

i 1 j 0 k $-\frac{\sqrt{2}}{2}$ l $-\frac{\sqrt{2}}{2}$

m $-\frac{\sqrt{2}}{2}$ n 1 o $\frac{\sqrt{3}}{3}$ p $-\frac{\sqrt{3}}{3}$

10

a $-\frac{1}{2}$ b $-\frac{\sqrt{3}}{2}$ c $-\frac{1}{2}$ d $\frac{\sqrt{3}}{3}$

11

a $-\frac{\sqrt{3}}{4}$ b -2

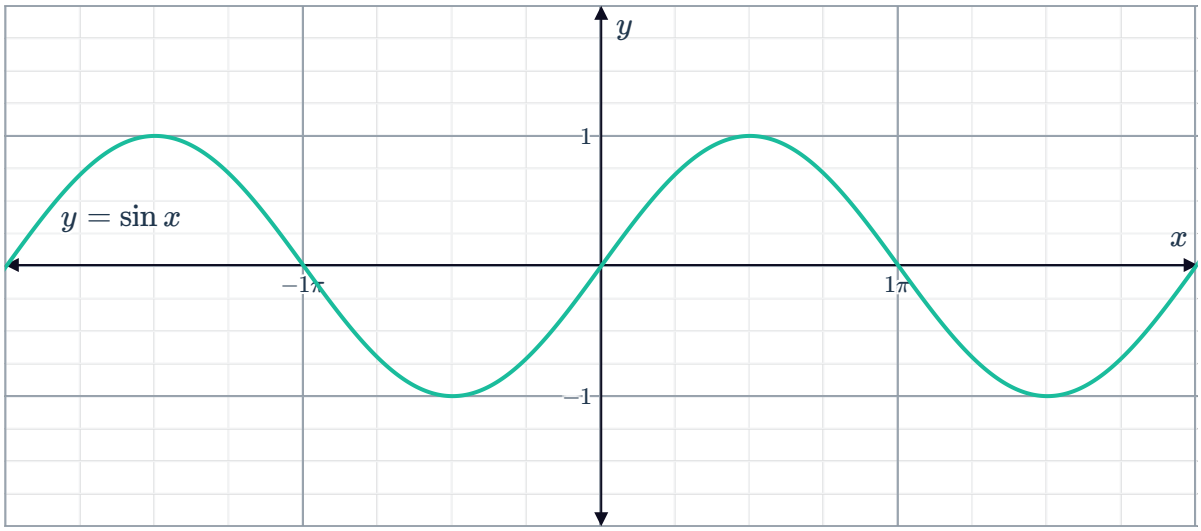
Trigonometric functions

12

a

x	0	$\frac{\pi}{6}$	$\frac{\pi}{2}$	$\frac{5\pi}{6}$	π	$\frac{7\pi}{6}$	$\frac{3\pi}{2}$	$\frac{11\pi}{6}$	2π
$\sin x$	0	$\frac{1}{2}$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	0

b



c 0

d Negative

e Negative

f Quadrant 3

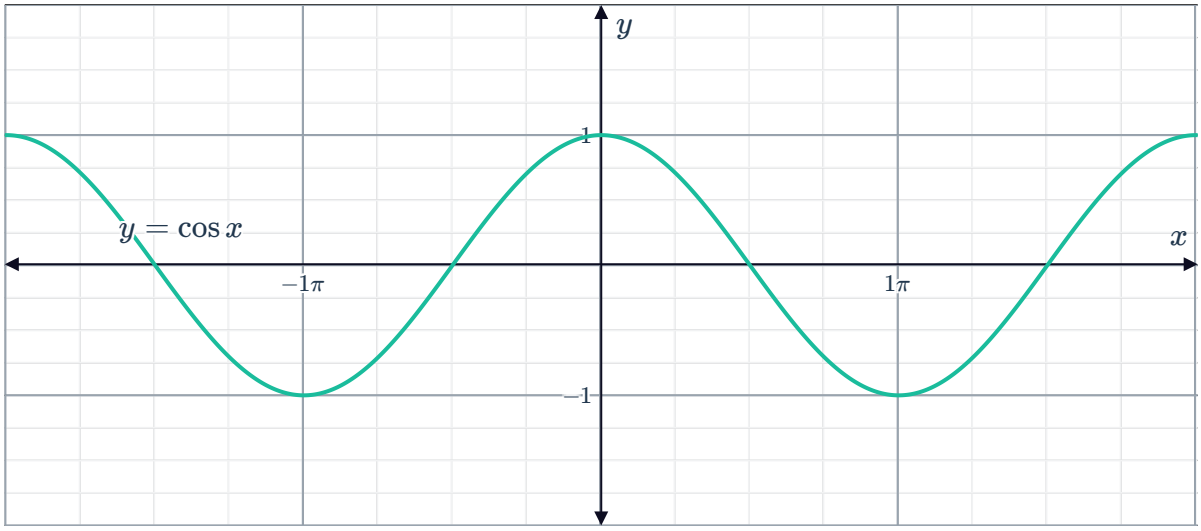
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13

a

x	0	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	2π
$\cos x$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1

b



c -1

d Positive

e Positive

f Quadrant 4

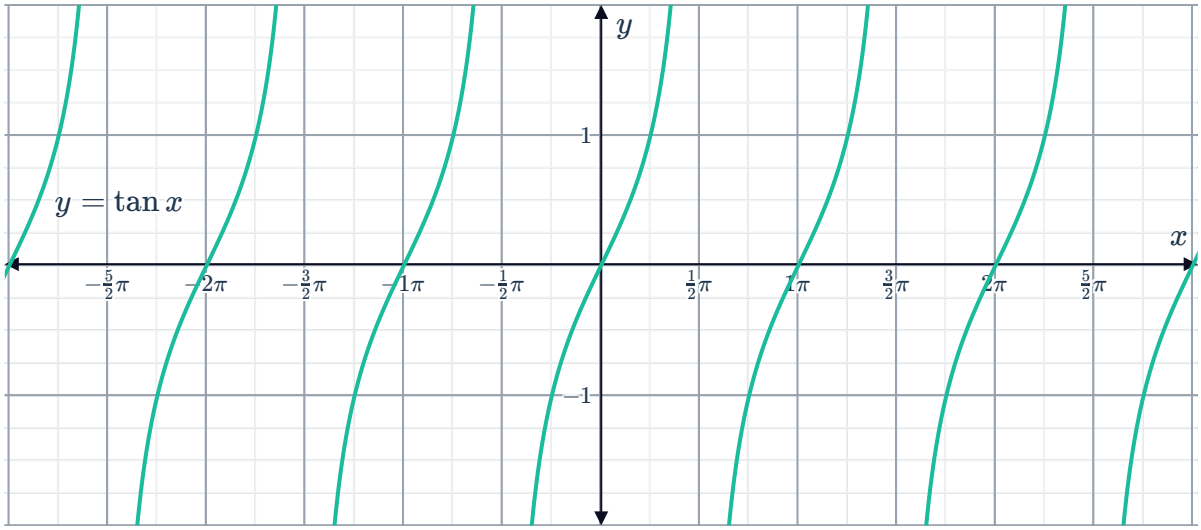
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14

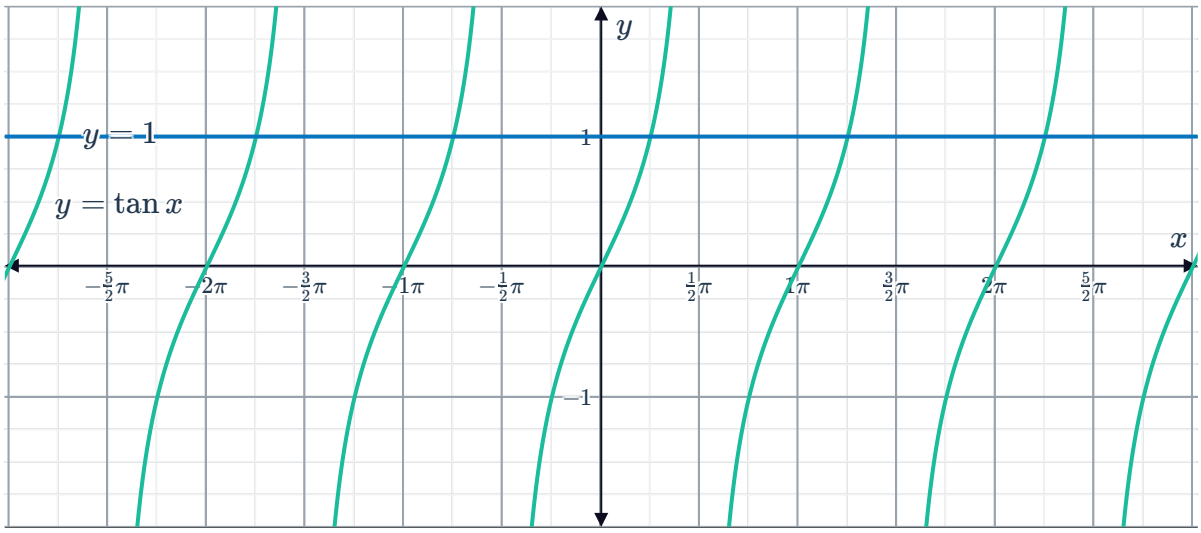
a

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\tan x$	0	1	undefined	-1	0	1	undefined	-1	0

b



c



d $x = -\frac{7\pi}{4}, -\frac{3\pi}{4}, \frac{\pi}{4}, \frac{5\pi}{4}$

e 0

f Negative

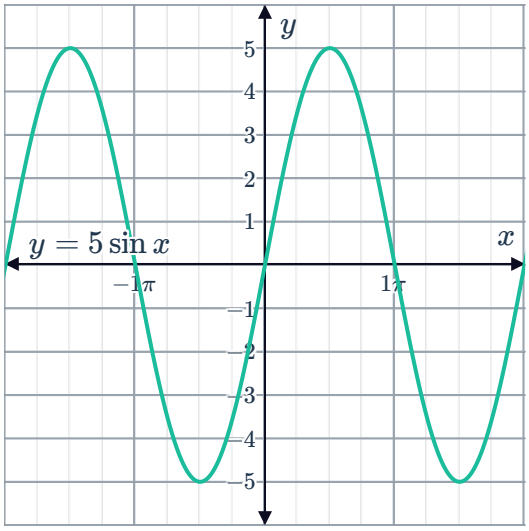
g Negative

h Quadrant 4

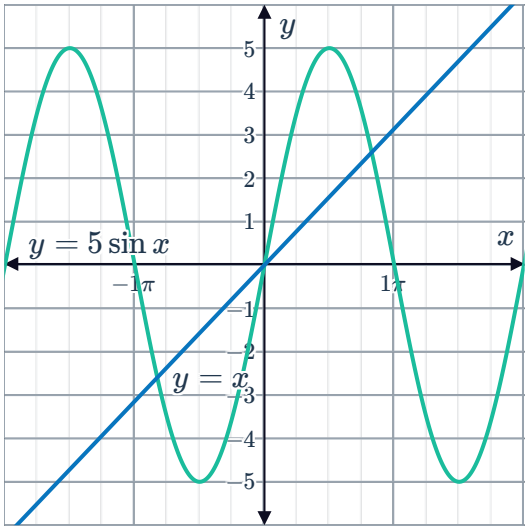
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a



b



c 3 solutions

16 Point C

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